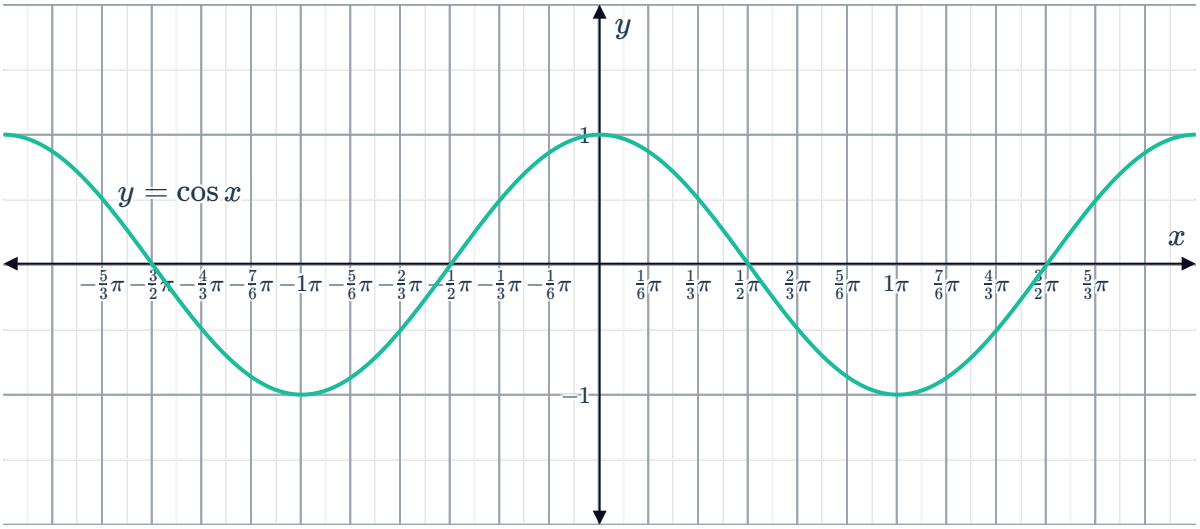
Secant function

1 Consider the graph of $y = \cos x$ for $-2\pi \leq x \leq 2\pi$:



a Complete the table of values, correct to three decimal places where necessary:

x	$-\frac{\pi}{4}$	0	$-\frac{\pi}{6}$	$\frac{2\pi}{3}$	π	$\frac{5\pi}{4}$
$\sec x$						

b Find the asymptotes of $y = \sec x$ in $-2\pi \leq x \leq 2\pi$.

c At what values of x is $\sec x = 1$?

d At what values of x is $\sec x = -1$?

e Find the period of $y = \sec x$.

f Sketch the graph of $y = \sec x$ over the domain $-2\pi \leq x \leq 2\pi$.

g For which intervals in the domain $-2\pi \leq x \leq 2\pi$ is the graph of $y = \sec x$ decreasing?

2 Consider the identity $\sec x = \dfrac{1}{\cos x}$.

a Complete the table of values, rounding each value to two decimal places:

x	57.30°	85.94°	89.38°	89.95°	90.53°	91.67°	114.59°
$\sec x$							

b What happens to the value of $\sec x$ when x approaches 90° from the left? Explain your answer.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

3 Consider the identity $\sec x = \dfrac{1}{\cos x}$.

a Complete the table of values, writing '-' if the value is undefined:

x	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\cos x$									
$\sec x$									

b State the minimum positive value of $\sec x$.

c State the maximum negative value of $\sec x$.

d On the same coordinate plane, sketch the graph of $y = \sec x$ and $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.

4 Consider the function $y = \sec x$:

a Sketch the graph of $y = \sec x$ for $-360^\circ \leq x \leq 360^\circ$.

b If $y = \sqrt{2}$ when $x = 45^\circ$, find the next positive x-value for which $y = \sqrt{2}$.

c Find the period of the graph.

d Find the smallest value of x greater than 360° for which $y = \sqrt{2}$.

e Find the first x-value less than 0° for which $y = \sqrt{2}$.

5 Consider the identity $\sec x = \dfrac{1}{\cos x}$ and the following table of values:

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\cos x$	1	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	1
$\sec x$									

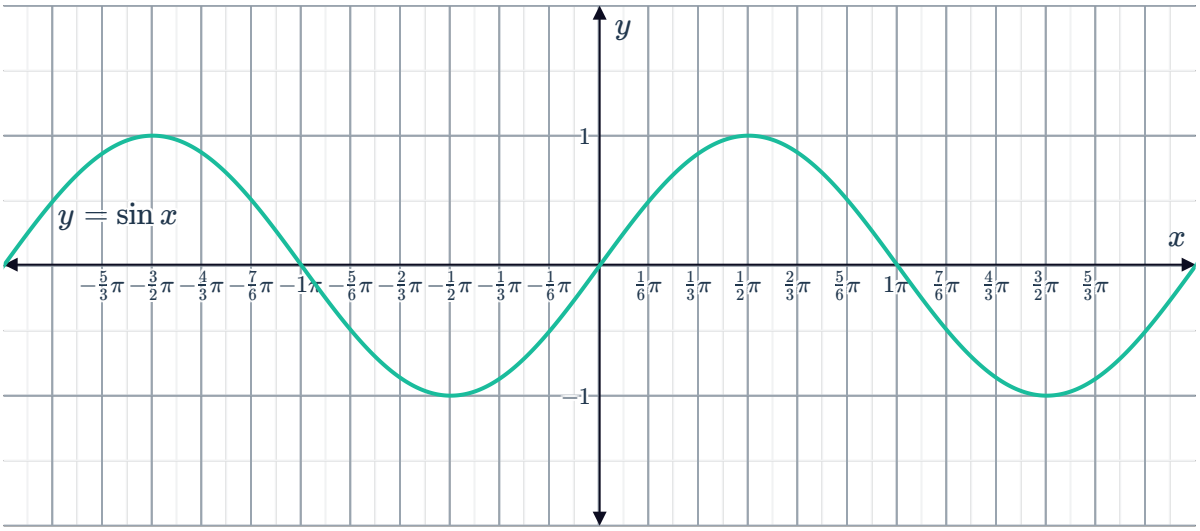
a Complete the table for the function $y = \sec x$.

b State the range for $y = \sec x$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Cosecant function

6 Consider the graph of $y = \sin x$ for $-2\pi \leq x \leq 2\pi$:



a Complete the table of values, correct to three decimal places where necessary:

x	$-\frac{2\pi}{3}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$
$\csc x$						

b State the asymptotes of $y = \csc x$ in $-2\pi \leq x \leq 2\pi$.

c At what values of x is $\csc x = 1$?

d At what values of x is $\csc x = -1$?

e Find the period of $y = \csc x$.

f Sketch the graph of $y = \csc x$ over the domain $-2\pi \leq x \leq 2\pi$.

g Find the y-intercept of the graph of $y = \csc x$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Consider the identity $\csc x = \dfrac{1}{\sin x}$.

a Complete the table of values, writing '-' if the value is undefined:

x	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\sin x$									
$\csc x$									

b State the minimum positive value of $\csc x$.

c State the maximum negative value of $\csc x$.

d Sketch the graph of $y = \csc x$ and $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$ on the same coordinate plane.

8 Consider the function $y = \csc x$:

a Sketch the graph of $y = \csc x$ for $-360^\circ \leq x \leq 360^\circ$.

b If $y = 2$ when $x = 30^\circ$, find the the next positive x-value for which $y = 2$.

c Find the period of the graph.

d Find the smallest value of x greater than 360° for which $y = 2$.

e Find the first x-value less than 0° for which $y = 2$.

9 Consider the identity $\csc x = \dfrac{1}{\sin x}$ and the following table of values:

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\sin x$	0	$\frac{1}{\sqrt{2}}$	1	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}}$	0
$\csc x$									

a State the values of x in the interval $[0, 2\pi]$ for which $\csc x$ is not defined.

b Complete the table for the function $y = \csc x$.

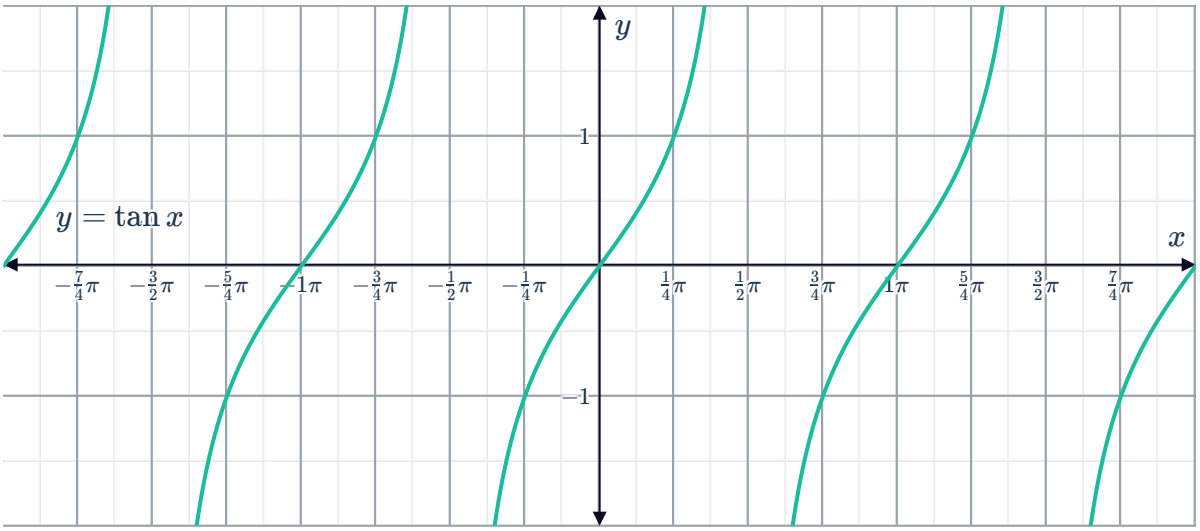
c State the range for $y = \csc x$.

d Sketch the graph of $y = \csc x$ on the interval $[-2\pi, 2\pi]$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Cotangent function

10 Consider the graph of $y = \tan x$ for $-2\pi \leq x \leq 2\pi$:



a Complete the table of values, correct to three decimal places where necessary:

x	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{4}$	$\frac{5\pi}{6}$
$\cot x$						

b Find the asymptotes of $y = \cot x$ in $-2\pi \leq x \leq 2\pi$.

c At what values would $\cot x = 0$?

d Find the period of $y = \cot x$.

e Sketch the graph of $y = \cot x$ over the domain $-2\pi \leq x \leq 2\pi$.

f Does the graph of $\cot x$ increase or decrease between any two successive vertical asymptotes?

11 Consider the graphs of $f(x) = \tan x$ and $g(x) = \cot x$.

a Describe how the asymptotes of one graph relate to the x-intercepts of the other.

b Explain how we could find the period of the functions by looking at the features of the graphs.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 Consider the identity $\cot x = \dfrac{\cos x}{\sin x}$.

a Complete the table of values, writing '-' if the value is undefined:

x	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\cot x$									

b Find the x-intercepts of the graph of $y = \cot x$ in the interval $[0^\circ, 360^\circ]$.

c Sketch the graph of $y = \cot x$ for $0^\circ \leq x \leq 360^\circ$.

13 Consider the function $y = \cot x$.

a Sketch the graph of $y = \cot x$ for $-360^{\circ\circ} \leq x \leq 360^\circ$.

b If $x = 30^\circ$, $y = \dfrac{1}{\sqrt{3}}$.Find the next positive x-value for which $y = \dfrac{1}{\sqrt{3}}$.

c Find the period of the graph.

d Find the smallest value of x greater than 360° for which $y = \dfrac{1}{\sqrt{3}}$.

e Find the first x-value less than 0° for which $y = \dfrac{1}{\sqrt{3}}$.

14 Consider the identity $\cot x = \dfrac{\cos x}{\sin x}$ and the following table of values:

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\cos x$	1	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	1
$\sin x$	0	$\frac{1}{\sqrt{2}}$	1	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}}$	0
$\cot x$									

a State the values of x in the interval $[0, 2\pi]$ for which $\cot x$ is not defined.

b Complete the table for the function $y = \cot x$.

c State the x-intercepts of the graph of $y = \cot x$ in the interval $[0, 2\pi]$.

d Sketch the graph of $y = \cot x$ on the interval $[-2\pi, 2\pi]$.

e Describe the graph of $y = \cot x$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647